

Computer Networks

Complex Computing Problem



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Topic: Efficient Subnetting Plan for University Network

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Relevant Material:  [Shared Folder Link](#)

Efficient IP Addressing and Subnetting Plan for University Network

Overview

I have to design an IP addressing and subnetting plan for a university with eight departments, using the 20.20.10.0/22 (1024 addresses). The goal is to use Variable Length Subnet Masking (VLSM) for efficient utilization, future scalability, and minimal wastage.

Department Size Estimation

Assuming typical university needs and future growth, we categorize departments as:

- Large:** Computer Science
- Medium:** Electrical Engineering, Mechanical Engineering, Civil Engineering, Library
- Small:** Admin Office, Hostel, faculty

Subnet Allocation Table

Department Name	Subnet ID	Subnet Mask	Broadcast Address	Host Bits	Total Hosts	Usable Hosts	Usable Range
Computer Science (CS)	20.20.8.0/24	255.255.255.0	20.20.8.255	8	256	254	20.20.8.1 – 20.20.8.254
Electrical Engg.	20.20.9.0/25	255.255.255.128	20.20.9.127	7	128	126	20.20.9.1 – 20.20.9.126
Library	20.20.9.128/25	255.255.255.128	20.20.9.255	7	128	126	20.20.9.129 – 20.20.9.254
Mechanical Engg	20.20.10.0/25	255.255.255.128	20.20.10.127	7	128	126	20.20.10.1 – 20.20.10.126
Civil Engg.	20.20.10.128/25	255.255.255.128	20.20.10.255	7	128	126	20.20.10.129 – 20.20.10.254
Hostel	20.20.11.0/26	255.255.255.192	20.20.11.63	6	64	62	20.20.11.1 – 20.20.11.62
Admin Office	20.20.11.64/26	255.255.255.192	20.20.11.127	6	64	62	20.20.11.65 – 20.20.11.126
Faculty Block	20.20.11.128/26	255.255.255.192	20.20.11.191	6	64	62	20.20.11.129 – 20.20.11.190
Free Space					64		20.20.11.191 – 20.20.11.255

Number of Devices and Future Growth

Large Subnet (Computer Science):

The Computer Science department likely hosts the most devices student and faculty computers, servers, lab equipment, and possibly research clusters. A larger subnet ensures enough IP addresses for all current devices and allows for future expansion without the need to readdress the network.

Medium Subnet (Engineering Departments, Library):

Departments like Electrical, Mechanical, Civil Engineering, and the library have a moderate number of devices enough to require more than a small subnet, but not as many as Computer Science.

Small Subnet (Admin Office, Hostel, Faculty):

These areas typically have fewer networked devices (e.g., office computers, printers, staff laptops).

Free space: The last /26 subnet (20.20.11.192/26) is left unassigned, explicitly available for future expansion or for a new department

Scalability: This approach ensures that if any department grows, the reserved subnet can be allocated, or subnets can be reorganized as needed

Justification (Why VLSM?)

VLSM allows assigning subnet sizes based on actual need, minimizing wastage and supporting future scalability¹.

Fixed-length subnetting would waste addresses, as all departments would receive the same size block regardless of need.

Summary of IPs Used and Remaining

Total Usable IPs Allocated: 254 (CS) + 126 (EE) + 126 (ME) + 126 (Civil) + 62 (Admin) + 126 (Library) + 62 (Faculty) + 62 (Hostel) = 944

Remaining: $2^6 = 64$

14 Ip's can be reserved for network infrastructure, routers, or future expansion

Total: $944 + 64 + 14 = 1024$

Challenges and Solutions

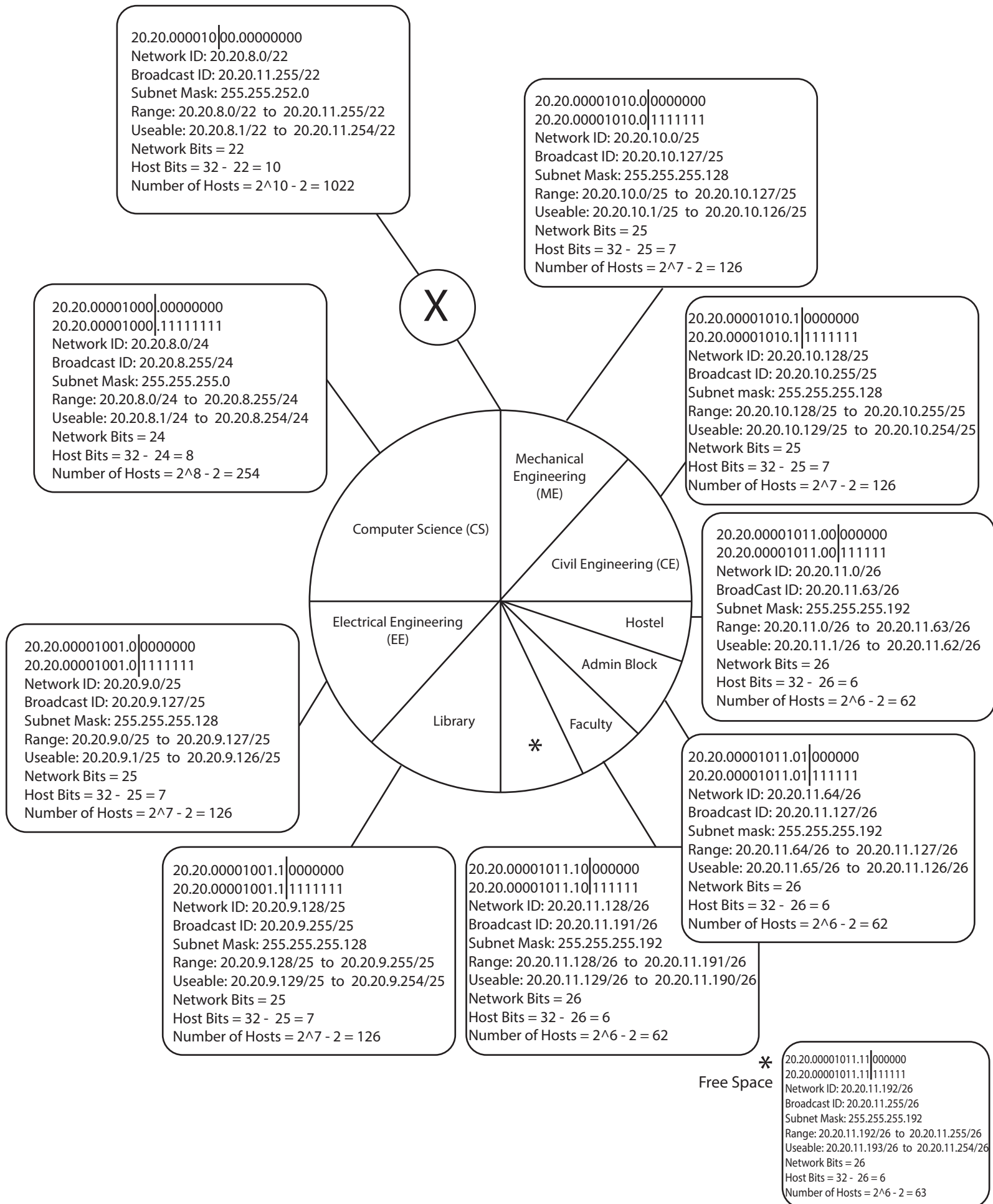
Challenge: If a department grows beyond its allocated subnet, reallocation may be necessary.

Solution: Plan for periodic review of IP usage and maintain documentation for easy reallocation.

Conclusion

This VLSM-based subnetting plan provides efficient, scalable, and well-documented IP address management for the university, meeting all objectives outlined in the assignment.

Subnetting Breakdown Diagram



Packet Tracer Implementation

